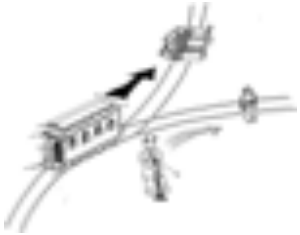
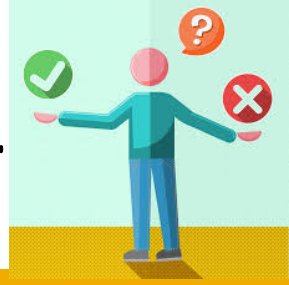
Please sit as close as possible to the front of the classroom – first four to five rows

Lots of people, a big classroom, and it could be difficult to hear



Global Engineering Ethics

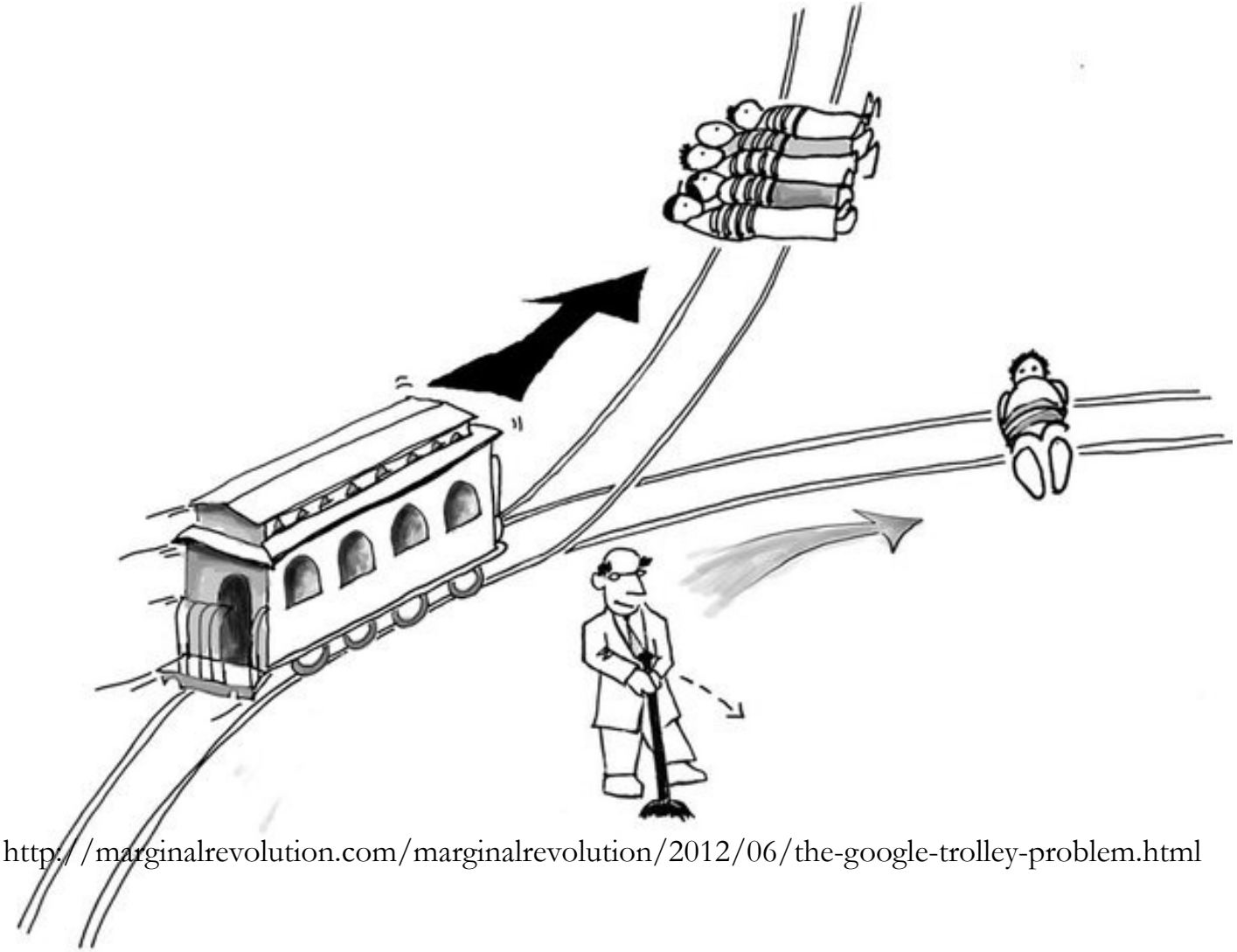
Thought experiments – ethical reasoning



What should be done?

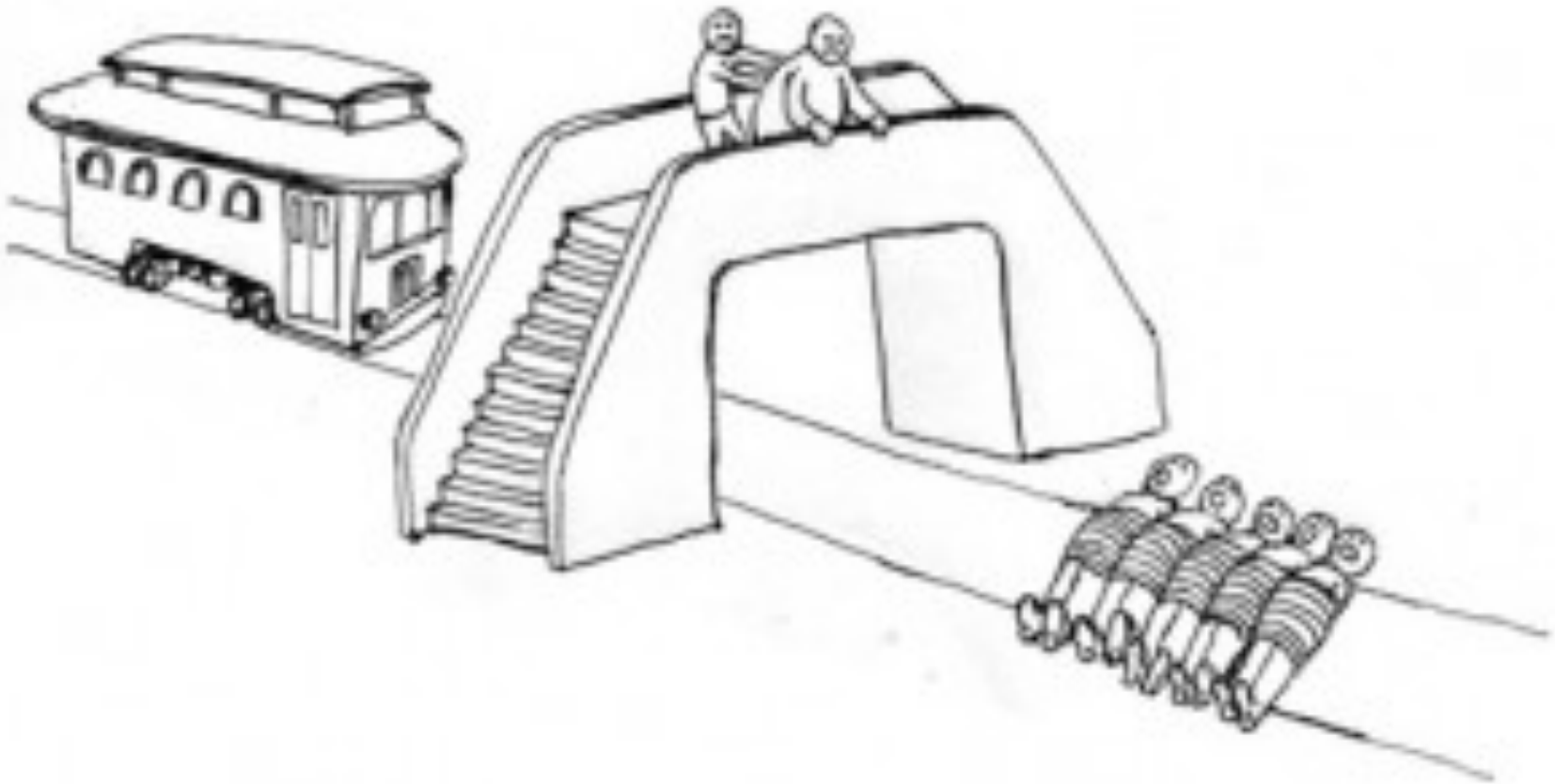
Why?

Trolley car dilemma



http://marginalrevolution.com/marginalrevolution/2012/06/the-google-trolley-problem.html

Footbridge dilemma



http://marginalrevolution.com/marginalrevolution/2012/06/the-google-trolley-problem.html

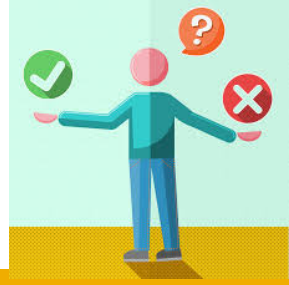
Class objectives

1. Evidence the ability to reason ethically

2. Explain why case-study analysis is used

3. Evidence the ability to complete steps one through five of the case-study procedure

Pair-share

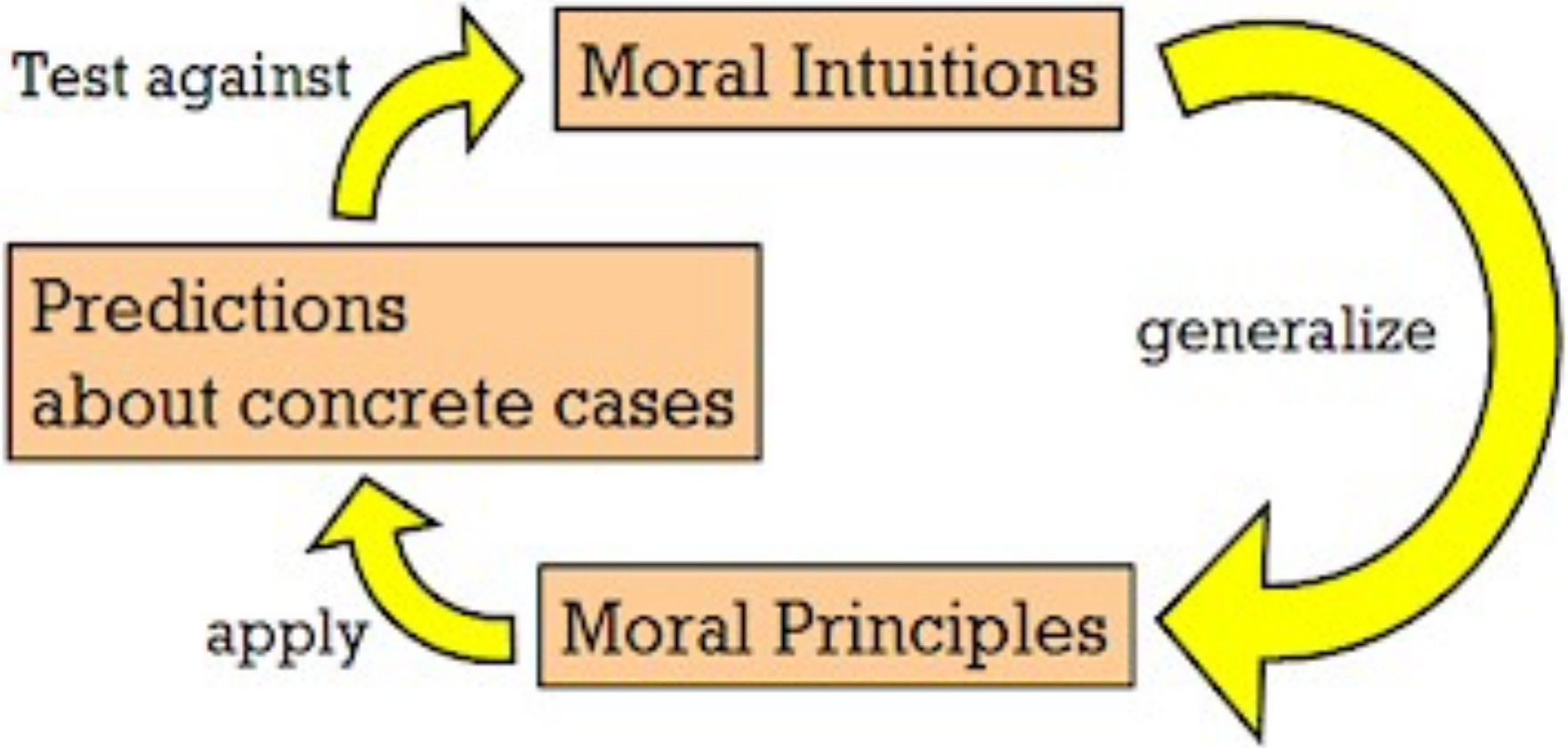


What should be done?

Why?

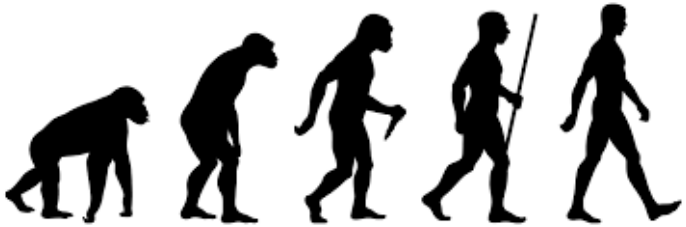
Thought experiments and reflective equilibrium

Thought experiments test our **theories** against **intuitions**, to confirm or modify them.



Review

- Ethics is about many things (pluralism) versus one (monism), and adaptive → universal + particular
- Unethical behaviors → ethical fading + competing goods
- Effects of technology and engineering huge, and diffuse in time and space → importance of but difficulties for ethics



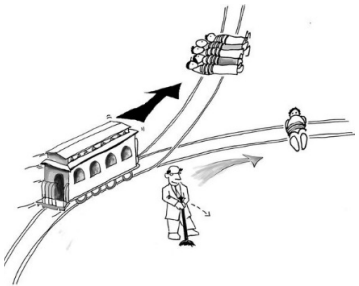
Why case-study analysis?

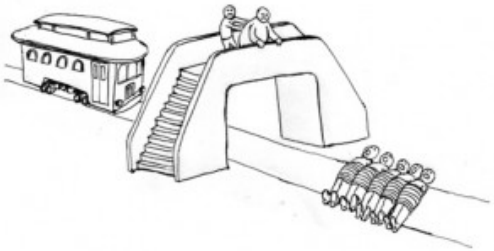
1. Contents of ethics within engineering
   - action + serious + others ← vague and broad
   - engineering ← technology ← novelty
2. Identifying (awareness) and resolving (reasoning) ethical dilemmas
   - step-by-step manner → reduces confusion
3. Hones critical thinking and facilitates active learning
   - engineering ~ ethical reasoning, for example, design specifications ← "better" and "worse," along a continuum (Whitbeck)

Thought experiments versus case-study

- Similarities = elicit intuitions, reflection, and assumptions, and tell us something about ethics, ethical judgments, and reasoning

- Differences = general versus specific, imagined versus real, simple versus complex





Case-study procedure

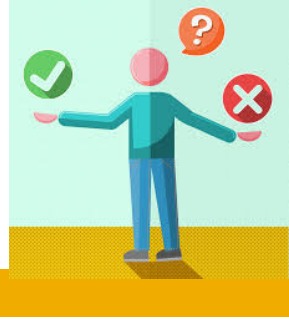
1. **Identifying ethical issues**
2. **Narrowing the focus**
3. **Determining relevant and missing facts**
4. **Making reasonable assumptions**
5. **Undertaking definitional clarification**
6. Conducting ethical analysis
7. Reviewing the process
8. Resolving the issue
9. Identifying practical constraints
10. Avoiding ethical problems

Building collapse – pair-share

Bill is the engineering manager at a large construction company. To do a favor for a friend, he instructs his subordinate, Bob, to hire his friend's daughter, Judy, as an engineering intern. Bob, in order to gain favor with his boss, assigns Judy to do the design calculations on a crucial part of an important building project. Three months after the completion of the project, part of the building collapses, killing three people and injuring several dozen more. An investigation by the regulatory agency determines that the problem was faulty calculations done by Judy.

1. Identify ethical issues
2. Narrow the focus
3. Identify relevant and missing facts
4. Make assumptions
5. Clarify terminology

Identifying ethical issues

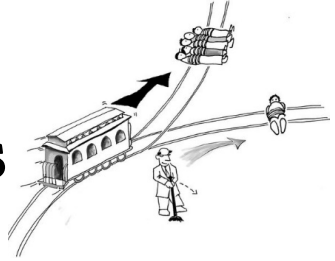


- Actions that have the potential to have a serious impact on the lives of other → Who or what is being affected, and how (=stakeholders)?

- Questions about what should or should not be done → Demand an answer, and show the world can be different

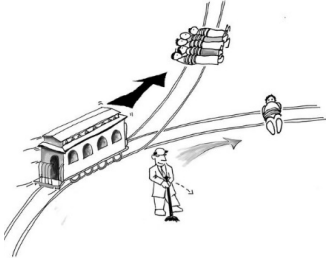
- Normative (right and wrong) versus factual (true and false)

Identifying ethical issues



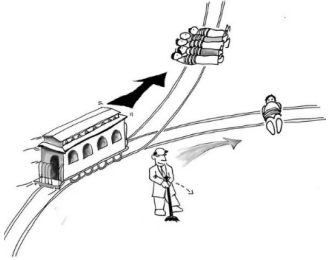
- **Should the man/you pull the switch?** If yes, then one person dies. If no, then five people die. Dying is seriously impactful, to oneself and others. Also, pulling or not pulling the switch could affect the man/oneself.
- **Should the man/you push the large man?**
- **Should Bob have assigned to Judy the building calculations?** Three people died and others were injured. In addition to these people, others would be seriously impacted, such as family and friends, those around the building, Judy, Bob, Bill, and other employees and stakeholders of the company, for instance, family members, and perhaps other firms in this industry and country.
- **Should Bill have instructed Bob to hire Judy?**
- **Who should be held responsible for the building collapse?**

Narrowing the focus



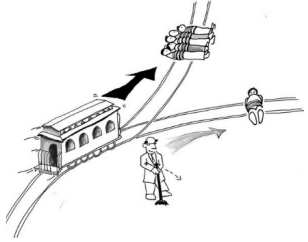
- Most serious
  - severity ~ worst consequences
  - scope ~ number affected
- Proximate (=direct) cause
- Root (=ultimate) cause

Narrowing the focus



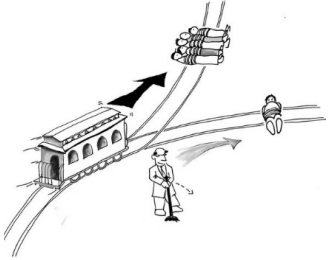
- Trolley car and footbridge dilemmas – only one each ← intentionally simplistic
- Should Bob have assigned…? It was the proximate cause of the collapse *
- **Should Bill have instructed…?** It was the root cause of the collapse *
- Who should be held responsible…? Backwards rather than forwards looking, and could be resolved with reference to the preceding two

Relevant and missing facts



1. Material facts: engineering dimensions of incidents
2. Individuals: what individuals did or didn't do
3. Organizations: persons-persons, persons-technology, etc., in terms of business, government, etc. policies and guidelines
4. Missing facts: information not present but relevant

Relevant and missing facts

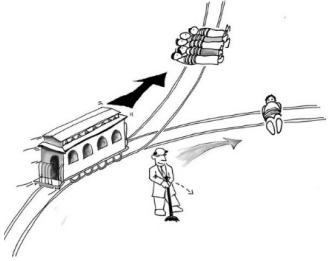


- Trolley car and footbridge dilemmas: If the man/you pull the switch/push the large man, then one person dies… Again, intentionally simplistic

1. A building collapsed, killing three people and injuring others.
2. Bill sought to do a favor for his friend. Bill directed Bob to hire Judy. Bob sought to court the favor of Bill. Bob assigned Judy to the building calculations.
3. The construction company is large. Bill is a manager. Bob is a subordinate of Bill. Judy is an intern.
4. Could Bill have refused the favor to his friend? How are hiring decisions normally made? Did Bill know Bob had assigned Judy the building calculations? Did anyone review Judy's work? Should someone have reviewed Judy's work – in other words, was there a policy in place? Was Judy qualified for the position?

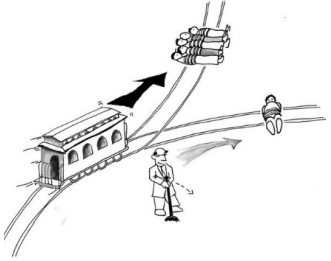
- Trolley car and footbridge dilemmas: Who're the people on the tracks, and why are they there?

Make assumptions



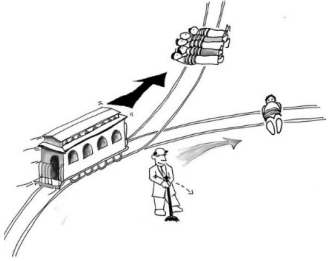
- Missing information ← common feature of real life
- Seek out additional information when necessary
- Justify this necessity
- Identify sources of disagreement
- Reasonable assumptions ← justified with regard to evidence

Make assumptions



- Probably, since the case study doesn't refer to any pressing necessity of Bill doing a favor for his friend, for instance, that his friend made him a loan, and there was some *quid pro quo*, which involved getting his daughter an internship.
- We don't know the answer to this question, but we might assume that, however these decisions are normally made, they would have prevented Judy from being hired. Otherwise, Bill wouldn't have had to do a favor and ask Bob to hire Judy.
- On the one hand, we know the company is large, so there's the possibility that Bill didn't know Bob has assigned Judy the calculations. On the other hand, we know Bob decided to assign Judy the calculations to court favor with Bill, so Bill would probably have known or found out since, otherwise, it wouldn't have made sense for Bob to assign Judy the calculations.
- Probably not, since, otherwise, the building would not have collapsed.
- Based on what we know about internships in companies, there was probably a policy in place requiring supervisors to review and certify Judy's work, especially if it was an important project.
- Probably not, otherwise, Bill wouldn't have had to ask Bob to hire Judy, and the building would not have collapsed.
- Trolley car and footbridge dilemmas: We're not supposed to know who the people on the tracks are, or why they're there, but this information might make a difference in our decision making, for example, women, younger versus older people, law-abiding versus -breaking citizens, etc.

Definitional clarification



- Clarity about words → precision in concepts
- Concepts can always be clarified
- Important since communication can fail, especially cross culturally and internationally.

Definitional clarification

- "others would be seriously impacted" = feel sad, lose money, other opportunities?
- "directed Bob to hire Judy" = asked or ordered?
- "court the favor" = to cause him to like, to like more, to get a promotion?
- "manager" = including engineering responsibilities?
- "review" = ensure the work was completed or completed correctly?
- "policy" = guidelines or required?
- "qualified" = previous courses, current course of study, degree, previous work experience?
- "large company" = number of people and organizational structure?

Summary

1. Ethical reasoning consists in the ability to make decisions about ethical issues – answering questions about what should or should not be done, giving reasons for one's answers.
2. Case-study analysis is used to familiarize students with the contents of ethics and ethical reasoning specific to technology and engineering, and ultimately to avoid unethical behaviors.
3. Steps one through five of the case-study procedure consist in identifying ethical issues, narrowing the focus, identifying relevant and missing facts, making assumptions, and clarifying terminology.

Next time

- Read chapter three and the case of McDonnell and Miller, and the ASME
- Complete chapter three reflection questions – do NOT need to complete the case-study procedure on McDonnell and Miller, and the ASME
- Begin thinking about who'll be in your group and your topics – best topics = ones related to engineering work you've done or experiences you've had – ethical or social implications of Vg100, Ve/m450, other projects and research, internship or work experiences you've had , or topics interesting to you